

# Targeting Vancomycin-Resistant *Enterococci* (VRE) Infections and Van Operon-Mediated Drug Resistance Using Dimeric Cholic Acid–Peptide Conjugates

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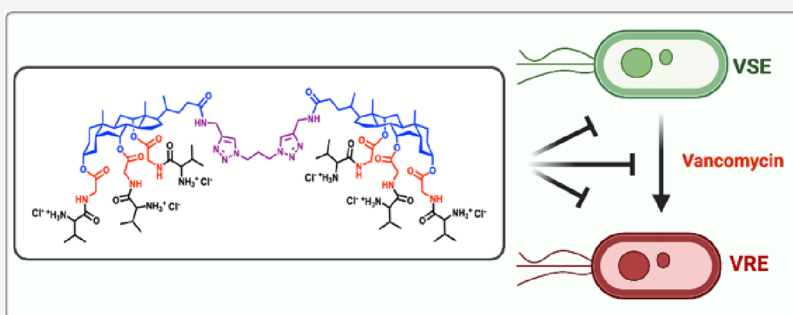
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**ABSTRACT:** Emergence of vancomycin resistance in Gram-positive bacteria and the prevalence of vancomycin-resistant *Enterococci* (VRE) infections are highly alarming as very limited antibiotic options are available against VRE infections. Here, we present the synthesis of cholic acid-derived dimeric amphiphiles where two cholic acid moieties are tethered through carboxyl terminals using different alkylene spacers. Our investigations revealed that dimer **5** possessing a propylene spacer and glycine-valine peptides tethered on hydroxyl groups is the most effective antimicrobial against VRE. Dimer **5** can permeabilize bacterial membranes, generate reactive oxygen species, and clear preformed biofilms. We further demonstrate that dimer **5** downregulates vancomycin-mediated transcriptional activation of the vanHAX gene cluster and does not allow VSE to develop vancomycin resistance until 100 generations. Therefore, this study, for the first time, presents a bacterial membrane-targeting amphiphile that can mitigate VRE infections and inhibit the emergence of vancomycin resistance.

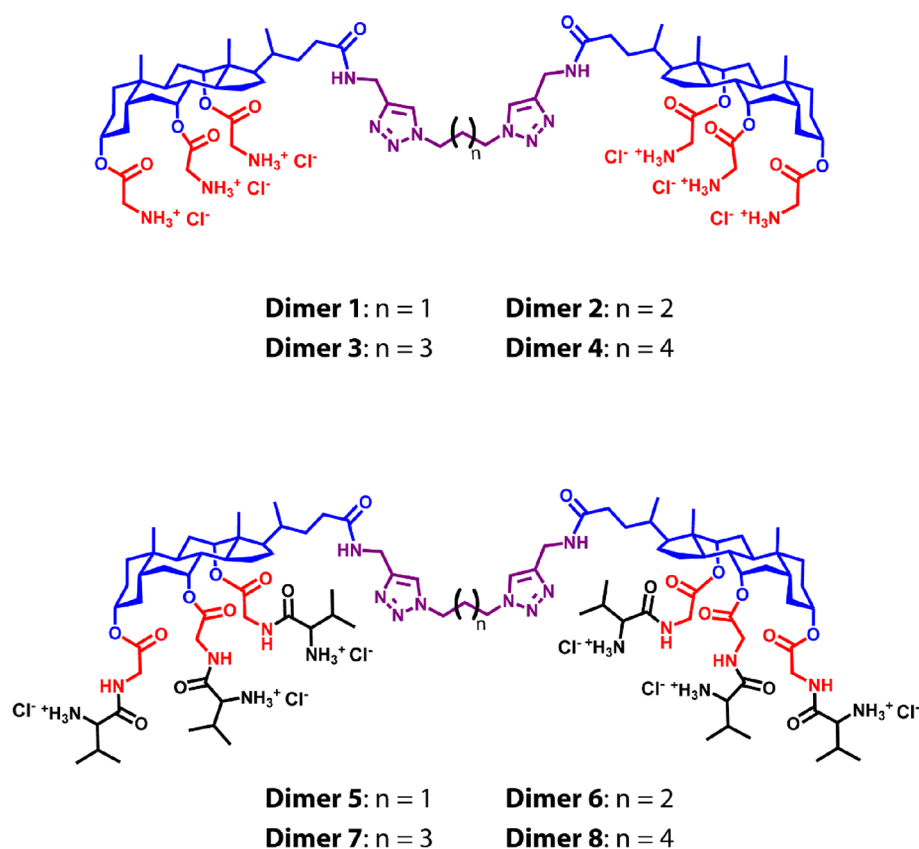
## 1. INTRODUCTION

*Enterococci* are opportunistic pathogens associated with lethal infections such as endocarditis, urinary tract infections, sepsis, wound infections, diabetic foot ulcers, and infections at surgical sites.<sup>1</sup> Glycopeptide antibiotics like vancomycin and teicoplanin A2-2 having glycosylated heptapeptides with tricyclic or tetracyclic nonribosomal peptide core are used as a last resort against these infections.<sup>2</sup> These antibiotics inhibit cell wall biosynthesis by binding to the D-alanyl-D-alanine (D-Ala-D-Ala) terminal sequence of precursor peptides and inhibit transpeptidation and transglycosylation in the peptidoglycan layer.<sup>3</sup> As glycopeptide antibiotics do not target any enzyme, bacteria find it difficult to develop resistance against these antibiotics.<sup>4</sup> However, the excessive use of antibiotics like vancomycin has allowed the emergence of numerous drug-resistant strains like vancomycin-intermediate *Staphylococcus aureus* (VISA), vancomycin-resistant *S. aureus* (VRSA), and vancomycin-resistant *Enterococci* (VRE).<sup>4,5</sup>

Emergence of drug resistance to vancomycin is caused by target site modification from D-Ala-D-Ala to either D-alanyl-D-serine (D-Ala-D-Ser) or D-alanyl-D-lactate (D-Ala-D-Lac) by the bacteria.<sup>6,7</sup> This target site modification is caused by Van operons, clusters of genes, that produce enzymes responsible for target site modifications. Among 11 different operons identified in VRE, the most common vanA operon (vanHAX gene cluster) consists of three resistance genes, vanH, vanX, and vanA. The gene vanH encodes dehydrogenase responsible for reduction of pyruvate to D-lactate, vanX encodes dipeptidase responsible for the hydrolysis of D-Ala-D-Ala, and vanA encodes a ligase for the synthesis of D-Ala-D-Lac.<sup>7</sup> vanR/

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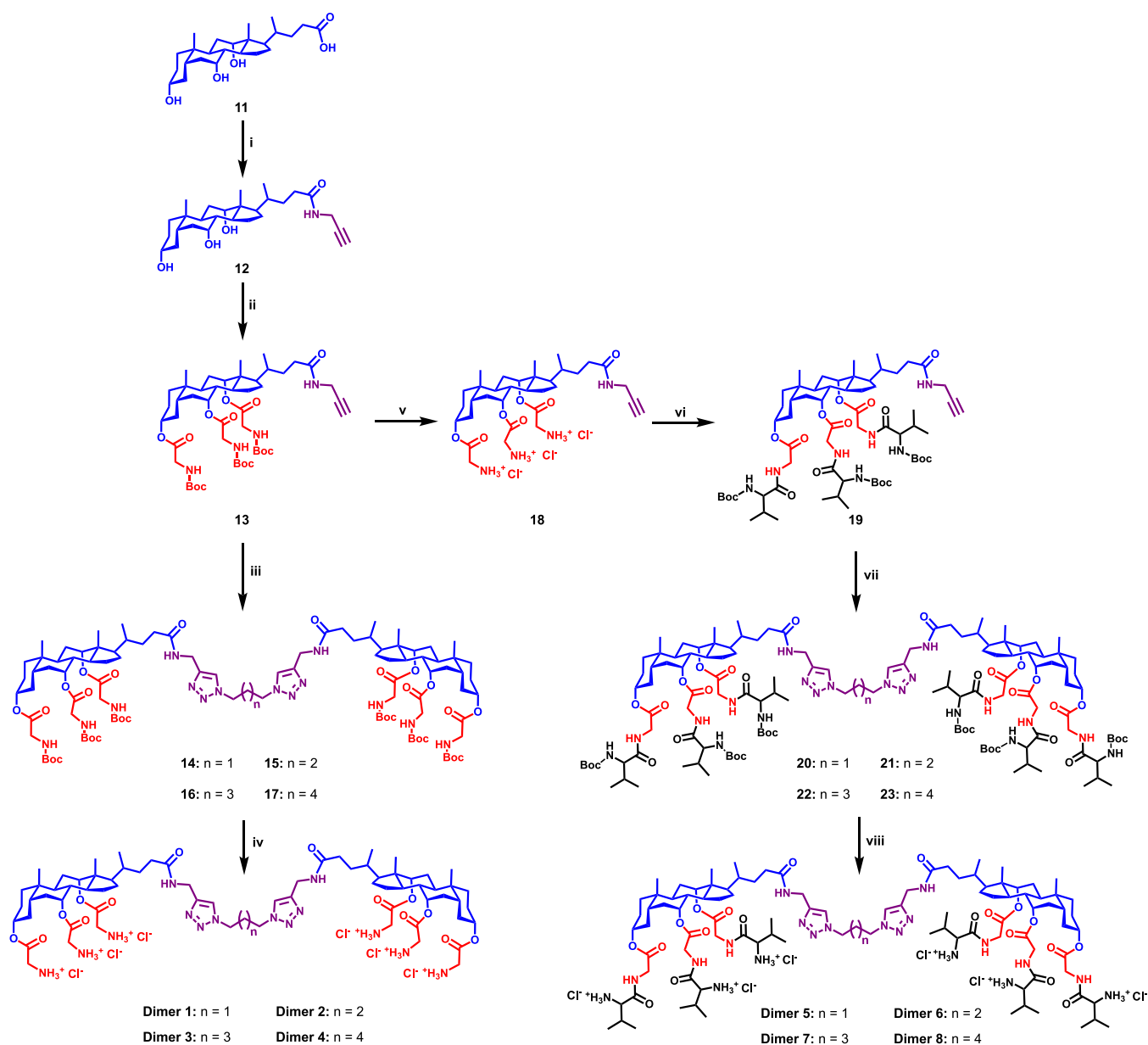
**Figure 1.** Schematic showing molecular structures of dimers of cholic acid–peptide conjugates.

S, a two-component system, regulates the expression of the vanHAX gene cluster. The presence of glycopeptide antibiotics causes autophosphorylation of membrane-associated VanS (a sensor kinase) followed by activation of a cytoplasmic response regulator and transcriptional activator, vanR protein, that is responsible for activation of the vanA operon.<sup>8</sup> Apart from activation of the vanA operon, the ability of *Enterococci* to form biofilms also allows them to develop non-inherited drug resistance.<sup>9</sup> An extra polymeric matrix released during biofilm formation reduces the penetration of antibiotics and results in the emergence of persistent infections.<sup>9</sup> As there are very limited options available for these VRE infections,<sup>10</sup> there is an urgent need for the development of strategies to antagonize vancomycin resistance.

There are three major strategies to antagonize vancomycin resistance where the first strategy involves the design of new antibiotics aiming novel targets in vancomycin-resistant bacteria.<sup>11</sup> The second strategy involves the synthesis of vancomycin derivatives using different strategies like modifying the peptide core, enhancing the binding affinity to target peptides, developing homomeric multivalent analogues, or by incorporating membrane-disruptive properties.<sup>12–18</sup> The third strategy involves designing of inhibitors targeting vancomycin-resistant enzymes encoded by genes like vanR, vanS, vanH, vanX, and vanY that modify vancomycin targets.<sup>19,20</sup> Numerous attempts have been made to develop chemical inhibitors against vanR and vanS, and their target genes vanA, vanH, and vanX.<sup>21,22</sup> Liu's group explored surface-localized antimicrobial display (SLAY)-derived peptides and showed their ability to antagonize vancomycin resistance in *Enterococci* by targeting the vanA operon.<sup>23</sup> Peptide agents like dodecamer peptide E12 can also inhibit a vanR/S two-component system

by binding to vanR protein.<sup>22</sup> However, to the best of our knowledge, there is no study emphasizing the ability of membrane-targeting antimicrobials to target the vanR/S two-component system and antagonize the vancomycin resistance.

Natural antimicrobial peptides (AMPs) are facially amphiphilic in nature and play a major role in lowering bacterial and viral infections.<sup>24</sup> Therefore, numerous small molecule- and polymer-based mimics of AMPs have been engineered for antimicrobial therapy.<sup>25</sup> These mimics of AMPs can bind to negatively charged phospholipids and disrupt the bacterial membranes or can disrupt the bacterial metabolism upon internalization.<sup>26</sup> Bile acid-derived amphiphiles due to their inherent facially amphiphilic character can easily mimic AMPs as they possess a cationically charged surface on one side and a hydrophobic domain on another side.<sup>27</sup> Cholic acid is a naturally occurring bile acid with facial amphiphilic character, and cholic acid-derived amphiphiles have been shown to act as effective antimicrobials.<sup>28–31</sup> Here, we present the synthesis of a new class of cholic acid-derived amphiphiles that can mitigate vancomycin-resistant *Enterococci* (Figure 1). These amphiphiles are highly selective in killing VRE bacteria by causing membrane depolarization followed by membrane disruptions. The most effective dimer 5 possessing a propylene spacer and glycine-valine peptides tethered on hydroxyl groups can also degrade drug-resistant biofilms and mitigate VRE-mediated wound infections. We further show that cholic acid-derived dimer 5 does not allow vancomycin-sensitive *Enterococci* to develop resistance against vancomycin until 100 generations by inhibiting the activation of the vanA operon.



**Figure 2.** Schematic showing the synthesis of dimers 1–4 and 5–8. Reagents and reaction conditions: (i) propargyl amine, EDC·HCl, HOBt, DIPEA, DMF, RT, overnight; (ii) Boc-glycine, DCC, DMAP, DCM, RT, overnight; (iii) different diazido alkanes, CuBr, TEA, MeOH, RT, 48–72 h; (iv) 4 M HCl in dioxane:DCM (1:5), 0 °C to RT, 3 h; (v) 4 M HCl in dioxane:DCM (1:5), 0 °C to RT, 3 h; (vi) Boc-valine, EDC·HCl, HOBt, DCM, RT, overnight; (vii) different diazido alkanes, CuBr, TEA, MeOH, RT, 48–72 h; (viii) 4 M HCl in dioxane:DCM (1:5), 0 °C to RT, 3 h.

## 2. RESULTS AND DISCUSSION

**2.1. Rationale and Design of the Study.** In our earlier study, we presented the synthesis of cholic acid–peptide conjugates where we conjugated dipeptides (glycine-X) using all 20 amino acids on three hydroxyl groups of cholic acid.<sup>31</sup> Our SAR studies showed that an amphiphile bearing glycine-valine is the most effective antimicrobial. Therefore, we selected glycine-valine dipeptides to modify the hydroxyl groups of cholic acid and also used only glycine modifications as a control. We designed and synthesized two series of amphiphiles. In first series, we conjugated three glycine moieties to cholic acid and synthesized four dimers 1–4 where two cholic acid–glycine conjugates were tethered through different alkylene spacers between them (Figure 1). In the second series of dimers 5–8, we used a cholic acid–

peptide conjugate where three glycine-valine dipeptides are tethered at three hydroxyl groups, and two of these cholic acid–peptide conjugates were further conjugated to each other through different alkylene spacers (Figure 1).

**2.2. Synthesis of Cholic Acid-Derived Dimers.** We prepared diazido alkanes (10a–10d) on reaction of dibromo alkanes (9a–9d) with sodium azide (Figure S1). For the synthesis of dimers 1–4, we conjugated propargyl amine to the C24 carboxyl group of cholic acid (compound 11) using EDC/HOBt coupling to get compound 12 (Figure 2). Compound 12 was then reacted with Boc-protected glycine using DCC/DMAP coupling to get compound 13 having three Boc-protected glycine moieties tethered at C3, C7, and C12 hydroxyl groups of cholic acid. Compound 13 was then reacted with different diazidoalkanes in the presence of Cu(I) to get Boc-protected dimers 14–17. Deprotection of Boc in the

**Table 1.** Table Showing the Antibacterial Activity of Dimer Molecules against Different Bacterial Strains along with Their Cytotoxic Profile against LLC Cells and Hemolytic Activity

|               | drug-susceptible strains<br>(MIC <sub>99</sub> , μg/mL) <sup>a</sup> |                   | drug-resistant strains<br>(MIC <sub>99</sub> , μg/mL) <sup>a</sup> |      | cytotoxicity (IC <sub>50</sub> , μg/mL) <sup>b</sup> | hemolytic activity (% at 50 μg/mL) <sup>c</sup> |
|---------------|----------------------------------------------------------------------|-------------------|--------------------------------------------------------------------|------|------------------------------------------------------|-------------------------------------------------|
|               | <i>S. aureus</i>                                                     | <i>E. faecium</i> | MRSA                                                               | VRE  | LLC                                                  | human RBCs                                      |
| dimer 1       | 128                                                                  | 32                | 32                                                                 | 16   | 131 ± 8                                              | 9                                               |
| dimer 2       | 128                                                                  | 64                | 64                                                                 | 32   | 211 ± 5                                              | 2                                               |
| dimer 3       | 128                                                                  | 32                | 64                                                                 | 16   | 153 ± 6                                              | 10                                              |
| dimer 4       | 128                                                                  | 16                | 32                                                                 | 8    | 66 ± 7                                               | 58                                              |
| dimer 5       | 64                                                                   | 16                | 16                                                                 | 4    | 414 ± 6                                              | 5                                               |
| dimer 6       | 128                                                                  | 8                 | 32                                                                 | 4    | 52 ± 7                                               | 50                                              |
| dimer 7       | 128                                                                  | 16                | 32                                                                 | 4    | 96 ± 3                                               | 36                                              |
| dimer 8       | 128                                                                  | 16                | 32                                                                 | 4    | 69 ± 8                                               | 57                                              |
| CAP 3         | 16                                                                   | 8                 | 4                                                                  | 2    | 40 ± 3                                               | 95                                              |
| vancomycin    | 8                                                                    | 4                 | 2                                                                  | >128 | >1 mM                                                | N.D.                                            |
| neomycin      | 16                                                                   | 16                | >128                                                               | >128 | >1 mM                                                | N.D.                                            |
| ciprofloxacin | <0.5                                                                 | 4                 | <0.5                                                               | 4    | N.D.                                                 | N.D.                                            |
| polymyxin B   | 128                                                                  | >128              | 32                                                                 | >128 | N.D.                                                 | N.D.                                            |

<sup>a</sup>MIC<sub>99</sub> (minimum inhibitory concentration) is the concentration that inhibits the growth of ~99% of bacterial population. <sup>b</sup>IC<sub>50</sub> is the concentration required to cause 50% cell death in mammalian cells (Lewis lung carcinoma cells). <sup>c</sup>Hemolytic activity (%) at the 50 μg/mL concentration. N.D.: not determined.

presence of acid conditions provided final dimers 1–4 (Figure 2). For the synthesis of dimers 5–8, Boc was deprotected from compound 13 to get compound 18, and Boc-protected valine was tethered to three glycine moieties of compound 18 using the EDC/HOBt coupling method to get compound 19 (Figure 2). The reaction of compound 19 with different diazidoalkanes in the presence of Cu(I) provided Boc-protected dimers 20–23. These dimers were then deprotected using 4 M HCl in dioxane to get final dimers 5–8 (Figure 2). All synthesized dimers were characterized by <sup>1</sup>H NMR, HRMS, and HPLC (Supporting Information).

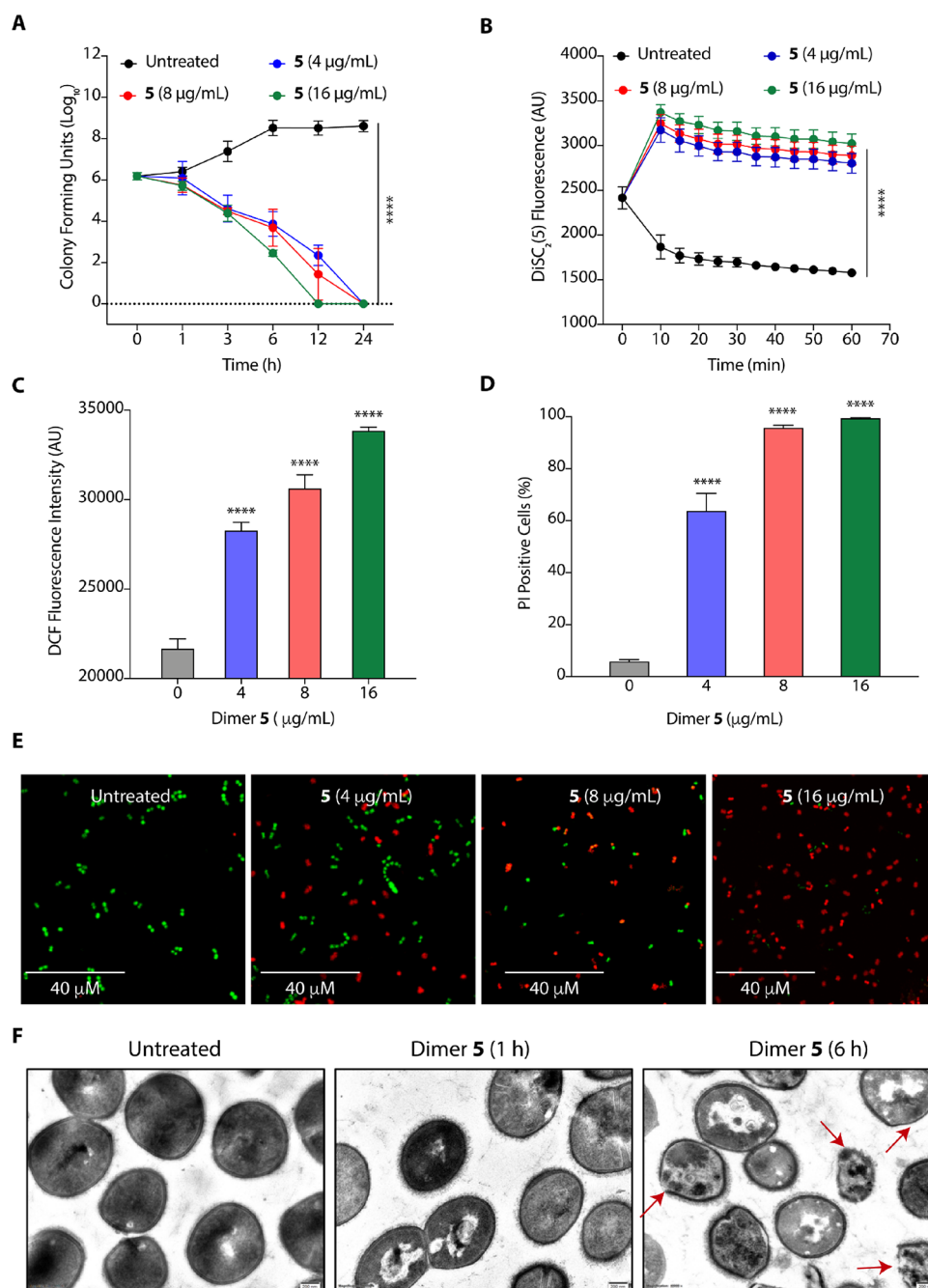
**2.3. Dimer 5 Is the Most Effective Antibacterial against VRE.** We tested the antibacterial activity of all the dimers against two different drug-sensitive strains, *S. aureus* and *Enterococcus faecium*, and two different drug-resistant strains, methicillin-resistant *S. aureus* (MRSA) and VRE, using broth dilution assay.<sup>32</sup> Structure–activity relationship (SAR) studies among dimers 1–4 demonstrated that these dimers are more active against *E. faecium* over *S. aureus* and are more effective against VRE over VSE (Table 1). Among dimers 1–4, dimer 4 is most active against VRE with an MIC<sub>99</sub> of 8 μg/mL. SAR studies among dimers 5–8 revealed that these molecules are also more active against *E. faecium* over *S. aureus*, and VRE are found to be more sensitive to these molecules. All dimers 5–8 have an MIC<sub>99</sub> of 4 μg/mL against VRE, whereas the MIC<sub>99</sub> against VSE is 8–16 μg/mL (Table 1). These studies witnessed that the synthesized dimer molecules are more active against *E. faecium* than *S. aureus*, and the VRE strain is more sensitive to dimers 5–8 than VSE (Table 1). Therefore, conjugation of dipeptide glycine-valine to cholic acid dimers makes these molecules highly effective antibacterials than cholic acid dimers having only glycine moieties. Interestingly, we also observed a monomeric analog (CAP 3) of dimer 5 that was more effective against VRE than VSE with an MIC<sub>99</sub> of 2 μg/mL.

As the cytotoxicity of membrane-targeting amphiphiles is a major bottleneck for clinical applications, we tested the toxicity of these amphiphiles against mouse lung epithelial cells (LLC) and determined IC<sub>50</sub> (minimum inhibitory concentration at which 50% cell death is observed) for all amphiphiles (Table

1). Among all the dimers, dimer 5 is the least cytotoxic with IC<sub>50</sub> > 410 μg/mL. Although dimers 6–8 were equally effective against VRE as dimer 5, these amphiphiles were found to be more toxic with an IC<sub>50</sub> of 50–100 μg/mL (Table 1). A monomeric analog (CAP 3) of dimer 5 is also highly toxic with an IC<sub>50</sub> of ~40 μg/mL. Calculation of the therapeutic index (IC<sub>50</sub>/MIC<sub>99</sub>) demonstrated that dimer 5 is the most effective in its therapeutic efficacy over other dimers and its own monomeric analog (Table 1). We also determined the hemolytic activity (human RBC lysis) of all amphiphiles at 50 μg/mL using a standard protocol.<sup>31</sup> We found that dimer 5 is the least hemolytic against human RBCs among all dimers (Table 1). Therefore, we conducted our further studies with dimer 5 to understand its mechanism of action, its ability to degrade biofilms, and its impact on vancomycin resistance.

**2.4. Dimer 5 Is Bactericidal and Disrupts Bacterial Membranes.** To decipher the bactericidal nature of dimer 5, we first performed time-dependent kill kinetic assay by quantifying the colony forming units (CFUs) at successive time points after the treatment of VRE with different concentrations of dimer 5. There was complete killing of bacterial cells at 4, 8, and 16 μg/mL (1×, 2×, and 4× MIC<sub>99</sub>) of dimer 5 within 24 h (Figure 3A), thereby confirming the bactericidal nature of the dimer. To examine the membrane-disrupting antibacterial mechanism of dimer 5, we used a membrane potential-sensitive 3,3-diethylthiadicarbocyanine iodide dye [DiSC<sub>2</sub>(5)] that gets accumulated in bacterial membranes, leading to quenching of its fluorescence.<sup>33</sup> DiSC<sub>2</sub>(5)-stained VRE cells were treated with dimer 5 at different concentrations (4, 8, and 16 μg/mL), and the change in fluorescence intensity was measured. A sharp dose-dependent increase in DiSC<sub>2</sub>(5) fluorescence clearly reflected the ability of dimer 5 to disrupt bacterial membrane potential (Figure 3B). The release of intracellular reactive oxygen species (ROS) including hydroxyl (·OH) ions, hydrogen peroxide (H<sub>2</sub>O<sub>2</sub>), hydroxy radicals (·OH), superoxide anion (O<sub>2</sub><sup>·−</sup>), singlet oxygen (<sup>1</sup>O<sub>2</sub>), and peroxide (O<sub>2</sub><sup>·−</sup>) is related to loss of membrane integrity as these free radicals cause membrane lipid peroxidation, mitochondrial dysfunction, and protein damage.<sup>34</sup> To elucidate the effect of dimer 5 on ROS



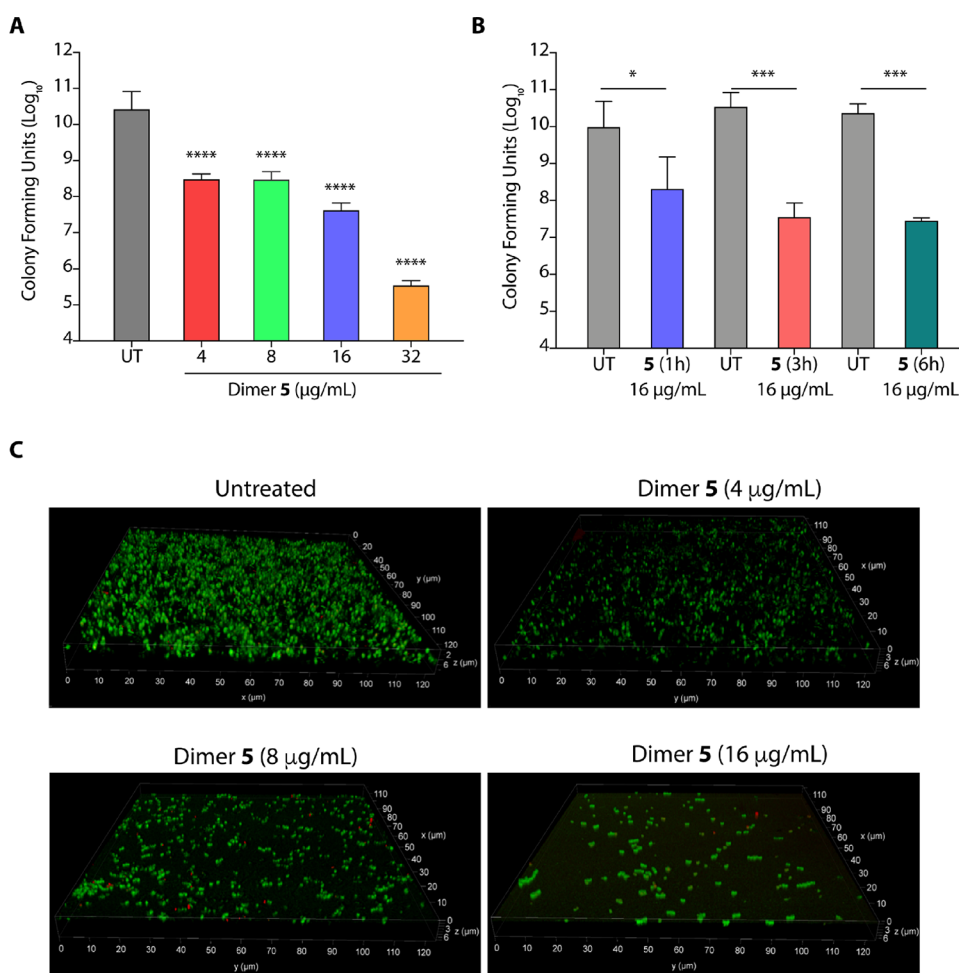


**Figure 3.** (A) Time-kill kinetic study showing the bactericidal effect of dimer **5** against VRE. (B) Increase in the fluorescence intensity of DiSC<sub>2</sub>(5) on the treatment of VRE with dimer **5** at different concentrations. (C) Increase in ROS on the treatment of VRE with different concentrations of dimer **5** for 1 h. (D) Percentage increase in the number of propidium iodide (PI)-positive VRE after 1 h of treatment of dimer **5**. (E) SYTO9/PI-stained fluorescence micrographs of untreated and dimer **5** (1×, 2×, and 4× MIC<sub>99</sub> for 1 h)-treated VRE. (F) Transmission electron micrographs after 1 and 6 h of treatment of VRE with dimer **5** (4 μg/mL). Data in panels (A) and (B) were analyzed using two-way ANOVA and those in panels (C) and (D) were analyzed using two-tailed unpaired Student's *t*-test. \*\*\*\**p* < 0.0001.

generation, we treated VRE cells with different concentrations of dimer **5**, and free ROS quantification was done using 2',7'-dichlorofluorescein diacetate (DCFH-DA) dye as it measures hydroxyl, peroxy, and other reactive oxygen species. We observed a dose-dependent increase in the fluorescence intensity of DCF in dimer **5**-treated VRE cells (Figure 3C), suggesting that dimer **5** enhances intracellular ROS production that further aids in its bactericidal activity.

Propidium bromide (PI) is a membrane-impermeable dye and would only stain the cells with compromised mem-

branes.<sup>35</sup> Therefore, we explored the ability of dimer **5** to permeabilize bacterial cells by measuring the percentage of PI-positive cells using flow cytometry. We observed ~60% PI-positive cells at 4 μg/mL (1× MIC<sub>99</sub>) treatment of dimer **5**, and treatment with 2× (8 μg/mL) and 4× (16 μg/mL) MIC<sub>99</sub> increased % PI-positive cells to 90–100% (Figure 3D). To visualize the damaged cells, we stained dimer **5**-treated cells with SYTO9 and PI as SYTO9 is a nucleic acid-binding cell membrane-permeable dye that gives fluorescence in the FITC region and PI fluorescence lies in the TRITC region.<sup>36</sup>



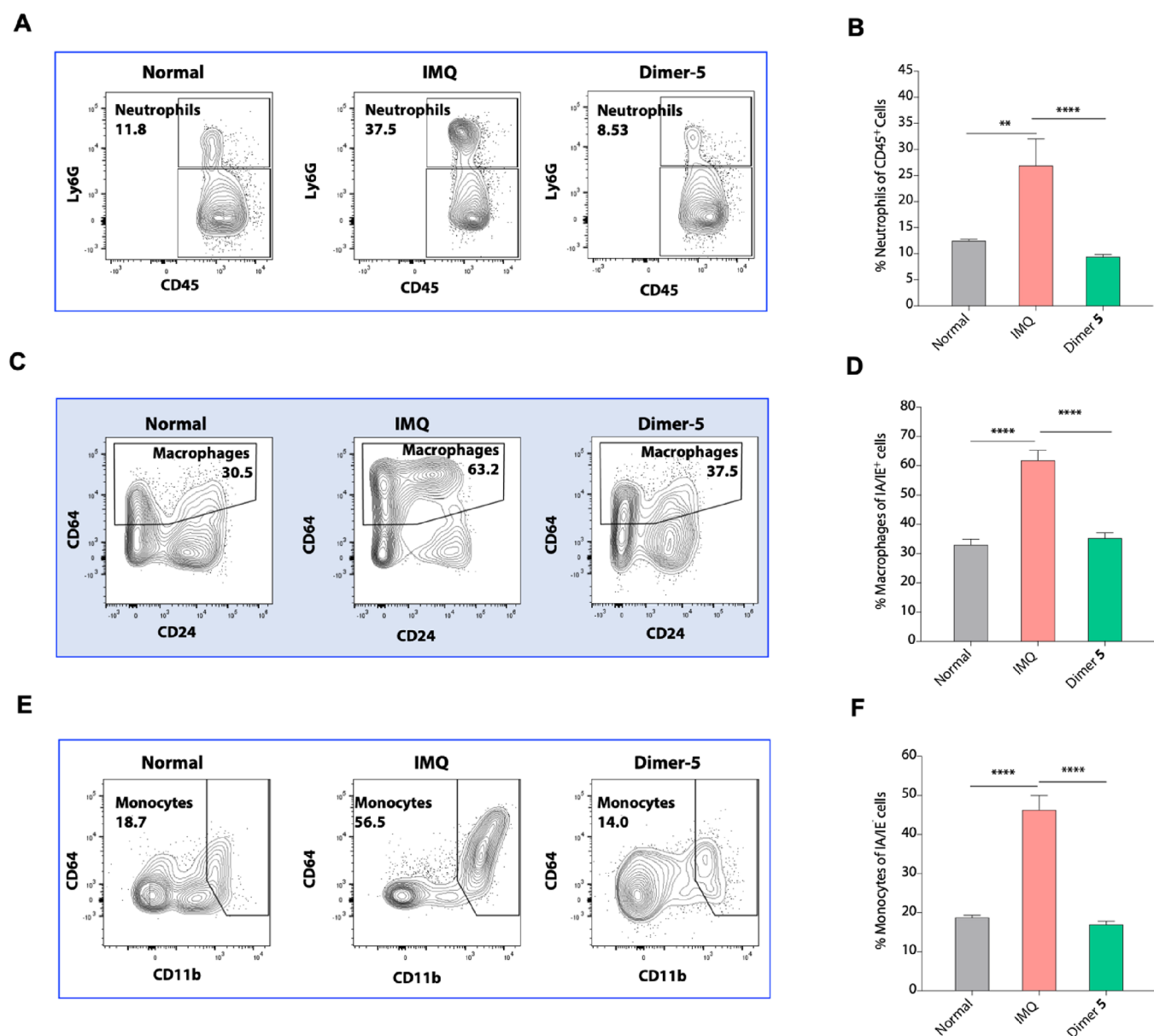
**Figure 4.** (A) Dose-dependent change in CFUs on the treatment of VRE biofilms for 12 h with dimer 5. (B) Time-dependent change in CFUs on the treatment of VRE biofilms with dimer 5. (C) Fluorescence micrographs of SYTO9/PI-stained untreated and dimer 5-treated VRE bacterial biofilms after 12 h of treatment with dimer 5. Data in panels (A) and (B) were analyzed using two-tailed unpaired Student's *t*-test. \**p* < 0.05, \*\*\**p* < 0.001, and \*\*\*\**p* < 0.0001.

Confocal images showing green and red fluorescence confirmed dimer 5-treated VRE cells with disrupted cell membranes where most of untreated cells had an intact membrane and got stained with SYTO9 only (Figure 3E and Figure S2). Transmission electron micrographs witnessed condensation of nucleic acids and membrane disruptions in a time-dependent manner, thereby demonstrating the ability of dimer 5 to disrupt bacterial membranes (Figure 3F and Figure S3).

**2.5. Dimer 5 Can Disrupt the VRE Biofilms.** Biofilms are multilayered structures of sessile form of microbes that have extracellular polysaccharide substances as the main component.<sup>37</sup> Bacterial cells protect themselves in these biofilms, restrict the antibacterial agents to penetrate, and limit their antibacterial activity.<sup>38</sup> VRE are known to form multidrug-resistant biofilms at fresh wound areas, indwelling devices, and medical implants. Therefore, we evaluated the antibiofilm activity of dimer 5, where VRE biofilms grown on coverslips were treated with different concentrations of the compound, and CFUs in biofilms were quantified. Dimer 5 was able to reduce CFUs in biofilms in a dose-dependent manner, and there was an ~3-log decrease in CFUs after treatment with 16 μg/mL dimer 5 and an ~5-log decrease in CFUs at 32 μg/mL dimer 5 (Figure 4A). To examine the time-dependent kinetics

of dimer 5 on VRE biofilms, CFUs were quantified at different time points (1, 3, and 6 h) and compared with untreated ones. Results revealed that dimer 5 can reduce the CFUs of biofilms by 3-log in comparison to untreated biofilms within 3 h of treatment (Figure 4B). We also stained the untreated and treated biofilms with SYTO9 and PI, and fluorescence micrographs confirmed that dimer 5 possessed antibiofilm activity as we observed a dose-dependent disruption of biofilms (Figure 4C). SYTO9/PI staining and CFU analysis also demonstrated that dimer 5 can also clear the *S. aureus* biofilms (Figure S4A,B). Therefore, these results confirm that dimer 5 can penetrate and disrupt the preformed VRE biofilms effectively.

**2.6. Dimer 5 Does Not Trigger Any Innate Immune Response.** Application of any foreign material is known to cause immune reactions through activation of Toll-like receptors that can further induce infiltration of immune cells.<sup>39</sup> This immune response to foreign material is initiated by infiltration of neutrophils, followed by monocytes and macrophages. Therefore, to find the clinical utility of dimer 5 for topical applications, we investigated the effect of dimer 5 on the innate immune system, where mouse skin was treated with dimer 5 three times a day for 6 consecutive days, and infiltration of neutrophils, monocytes, and macrophages was

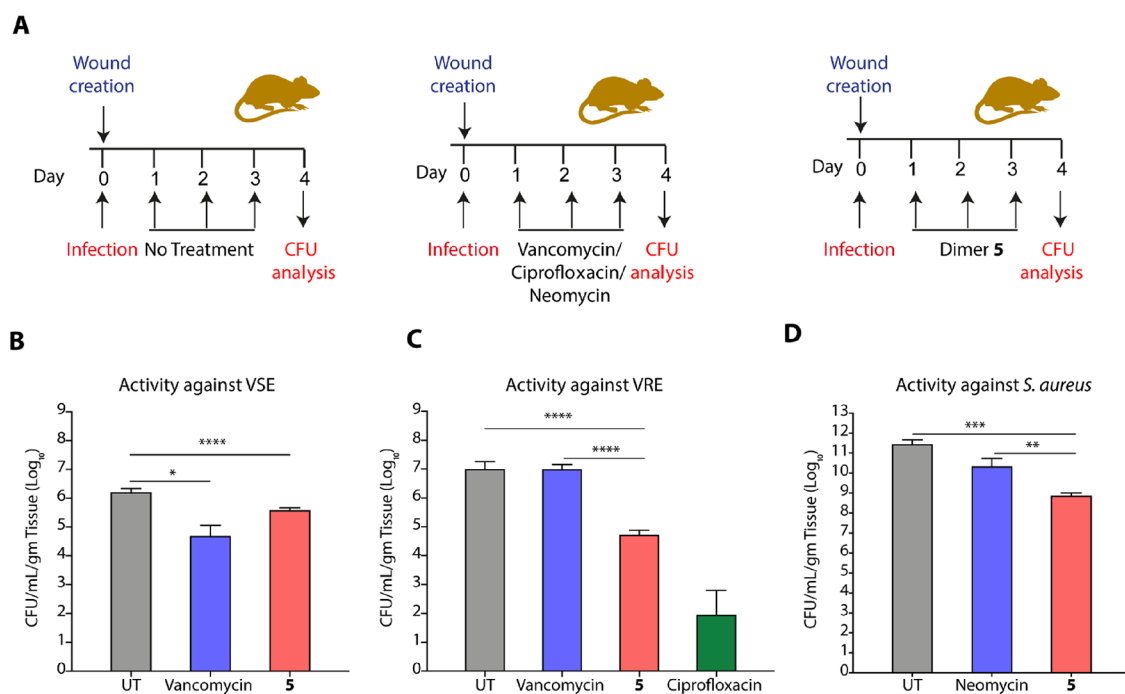


**Figure 5.** (A, B) Representative contour plots showing Ly6G-positive neutrophils on different treatments (A) and quantification of infiltrated neutrophils in untreated, IMQ-treated, and dimer 5-treated skin tissues (B). (C, D) Representative contour plots showing CD64-positive macrophages on different treatments (C) and quantification of infiltrated macrophages in untreated, IMQ-treated, and dimer 5-treated skin tissues (D). (E, F) Representative contour plots showing CD64 and CD11b double-positive monocytes (E) and quantification of infiltrated monocytes in untreated, IMQ-treated, and dimer 5-treated skin tissues. Data in panels (B), (D), and (F) were analyzed using two-tailed unpaired Student's *t*-test. \*\**p* < 0.01, \*\*\*\**p* < 0.0001.

quantified. We used imiquimod (IMQ) as a positive control that is known to induce inflammation by activating the Toll-like receptor 7 (TLR7).<sup>40</sup> A single-cell suspension of excised skin tissues from untreated, dimer 5-treated, and IMQ-treated mice was stained with cocktail of antibodies and analyzed by flow cytometry. We gated a single-cell population for live cells using DAPI, and live cells were further gated for CD45-positive leukocytes (Figure S5).<sup>41</sup> CD45-positive cells were further gated for Ly6G-positive neutrophils, and Ly6G-negative non-neutrophils were gated for MHC Class II (IA/IE)-positive cells that consist of monocytes and macrophages (Figure S5). For identification of macrophages, MHC Class II (IA/IE)-positive cells were gated for CD64-positive cells, and CD11b and

CD64 double-positive cells were assigned to monocytes (Figure S5).

Flow cytometry analysis showed a >2-fold increase in infiltration of neutrophils on application of IMQ, whereas we did not find any significant change in neutrophil counts on application of dimer 5 (Figure 5A,B). We also observed a >2-fold increase in accumulation of macrophages in inflamed skin samples on IMQ application and did not observe any macrophage infiltration in dimer 5-treated mice (Figure 5C,D). Similarly, application of dimer 5 did not cause any induction of monocytes to the treated skin tissues, whereas IMQ application caused a significant increase in infiltration of monocytes (Figure 5E,F). Therefore, these results confirm the



**Figure 6.** (A) Schematic showing the design of animal studies where dimer **5** is tested for its antibacterial activity against VSE, VRE, and *S. aureus*. (B, C) Change in the number of colonies of VSE (B) and VRE (C) bacteria treated topically with vancomycin and dimer **5** for three times a day for 3 days at a dose of 40 mg/kg. (D) Change in CFUs of *S. aureus* on topical treatment with neomycin and dimer **5** for three times a day for 3 days at a dose of 40 mg/kg. Data in (B)–(D) were analyzed using one-way ANOVA. \* $p < 0.05$ , \*\* $p < 0.01$ , \*\*\* $p < 0.001$ , \*\*\*\* $p < 0.0001$ .

non-immunogenic nature of dimer **5**, thereby making it suitable for topical applications.

**2.7. Dimer **5** Can Mitigate Skin Infections.** As VSE and VRE are one of the major reasons for wound and soft tissue infections,<sup>42</sup> we assessed the ability of dimer **5** to alleviate wound infections. We created an  $\sim 1$  cm<sup>2</sup> circular wound on BALB/c mice and infected the wounds with VSE or VRE cells. After 18 h of infection, we randomized the mice into three groups (four to five mice/group) with the first group being untreated mice (Figure 6A). Group 2 mice were treated with vancomycin, and group 3 mice were treated with dimer **5**. We also used ciprofloxacin as a positive control in VRE-infected mice. Topical treatments were continued for three times a day for 3 consecutive days. After treatment, we excised skin tissue, homogenized them, and quantified the colony-forming units. Quantification of CFUs in mice infected with VSE bacteria showed an  $\sim 2$ -log-fold decrease in CFUs on vancomycin treatment, and dimer **5** induced only an  $\sim 1$ -log-fold decrease in CFUs (Figure 6B). In contrast, we observed a 2-log decrease in CFUs on dimer **5** treatment in VRE-infected mice, and vancomycin could not induce any bactericidal effect (Figure 6C). These results clearly demonstrated that dimer **5** is more effective against VRE as compared to VSE and is able to mitigate VRE-mediated wound infections. We also validated the bactericidal effect of dimer **5** against *S. aureus*-mediated wound infections, where dimer **5** was able to cause a  $>2$ -log decrease in bacterial load and was more effective than neomycin (Figure 6D).

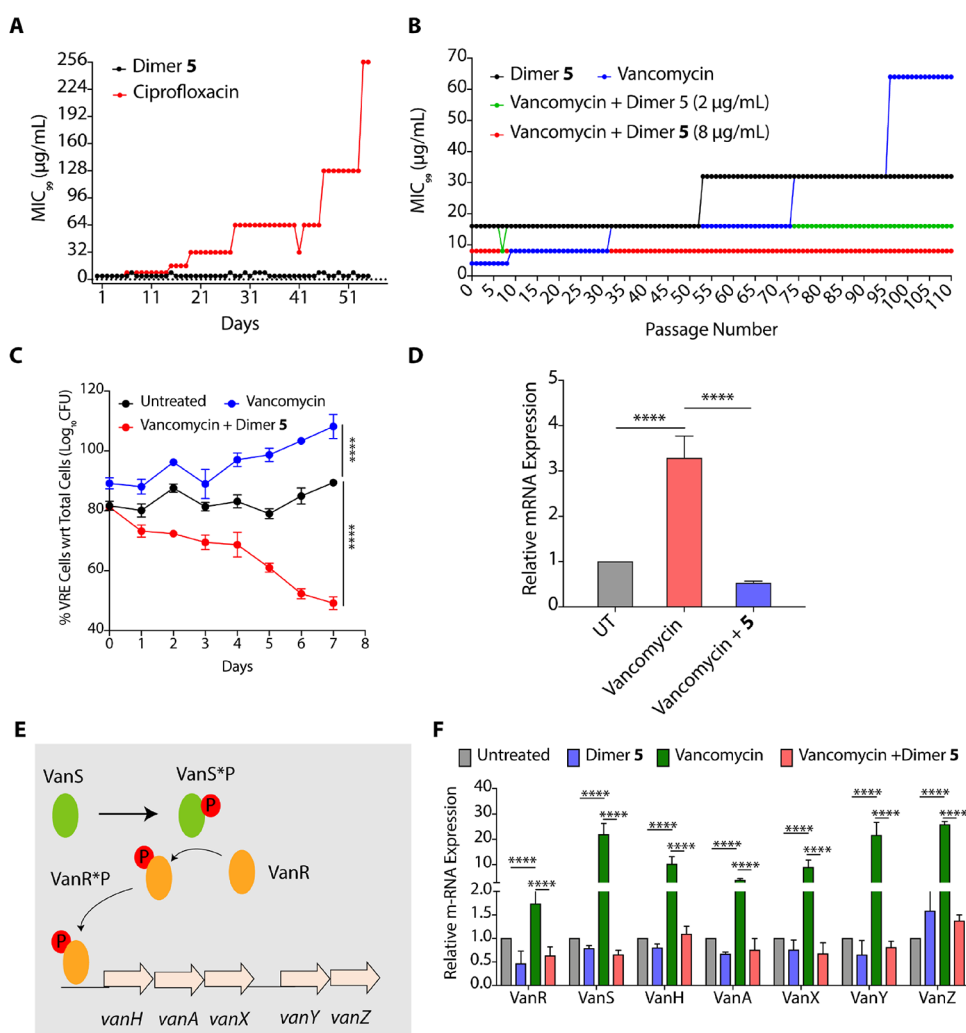
**2.8. Dimer **5** Can Antagonize Vancomycin Resistance.** As the continuous treatment of bacteria with antimicrobials allows them to develop drug resistance,<sup>43</sup> we tested the ability of VRE to develop drug resistance against dimer **5** and used ciprofloxacin as a positive control. Bacterial cells were treated with dimer **5** and ciprofloxacin to determine MIC<sub>99</sub>, and sub-

MIC<sub>99</sub>-treated cells were grown further and tested for MIC<sub>99</sub>. We did not observe any significant change in MIC<sub>99</sub> of dimer **5** against VRE, whereas there was a many-fold increase in MIC<sub>99</sub> of ciprofloxacin (Figure 7A). Therefore, these results confirm that VRE bacteria cannot develop resistance against dimer **5** that may be due to its ability to target bacterial membranes.

As the emergence of vancomycin resistance itself is a serious threat, we investigated the effect of vancomycin alone, dimer **5** alone, and a combination of vancomycin and dimer **5** on VSE bacteria and assessed the ability of VSE bacteria to develop resistance against these combinations. We observed that VSE started developing resistance to vancomycin as there was a 2-fold increase in MIC<sub>99</sub> after 10 generations, a 4-fold increase after 30 generations, and an 8-fold increase after 70 generations (Figure 7B). VSE bacteria showed only a 2-fold increase in MIC<sub>99</sub> after 50 generations against dimer **5**, and there was no further increase in MIC<sub>99</sub>. Interestingly, we observed that in the presence of dimer **5** (2  $\mu$ g/mL), VSE bacteria showed an MIC<sub>99</sub> of 16  $\mu$ g/mL against vancomycin, and there is no increase in MIC<sub>99</sub> on different passages until 110 generations (Figure 7B). Similarly, the presence of dimer **5** (8  $\mu$ g/mL) does not allow the VSE to develop resistance against vancomycin until 110 passages (Figure 7B). Therefore, these results clearly indicate that dimer **5** does not allow the emergence of vancomycin resistance in VSE.

To further evaluate the efficacy of dimer **5** to inhibit vancomycin resistance, we performed a competitive assay, where a mixture of VRE and VSE (1:1) was continuously passaged at vancomycin, dimer **5**, and a combination of vancomycin and dimer **5**. We used sub-inhibitory concentrations of 2  $\mu$ g/mL for dimer **5** (MIC<sub>99</sub> = 4  $\mu$ g/mL against VRE) and 2  $\mu$ g/mL for vancomycin (MIC<sub>99</sub> = 4  $\mu$ g/mL against VSE). The MIC<sub>99</sub> values of dimer **5** and vancomycin as a combination (VRE + VSE) are 16 and  $>128$   $\mu$ g/mL,



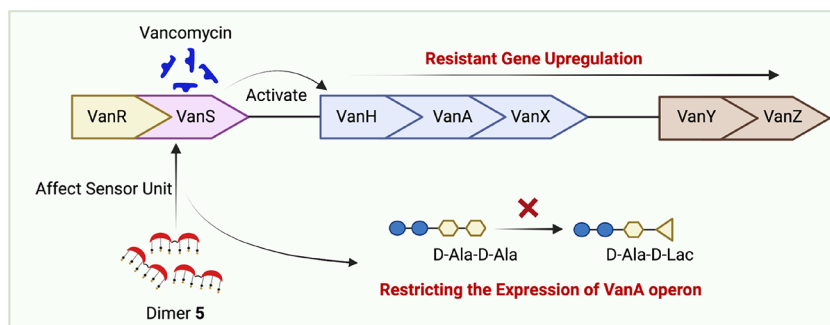


**Figure 7.** (A) Change in MIC<sub>90</sub> of VRE for dimer 5 and ciprofloxacin. (B) Generation-dependent change in MIC<sub>90</sub> of VSE on the continuous treatment with dimer 5 alone, vancomycin alone, or the combination of dimer 5 and vancomycin. (C) Percentage change in CFUs of VRE bacteria at different passages with respect to total cells after treatment. (D) Change in gene expression of the Tn1546 transposon element in different treatments groups at 7 day of coculture study. (E) Schematic showing the vanR/S two-component system regulating the gene expression responsible for vancomycin resistance. (F) Change in expression of vanA operon genes in VRE on treatment with dimer 5 alone, vancomycin alone, or the combination of vancomycin and dimer 5 for 1 h at sub-inhibitory concentrations. Data in panel (C) were analyzed using two-way ANOVA, and those in panels (D) and (F) were analyzed using two-tailed unpaired Student's *t*-test. \*\*\*\**p* < 0.0001.

respectively. The untreated and treated cells were plated on vancomycin (6 μg/mL) agar plates that allow only the growth of vancomycin-resistant cells. As expected, vancomycin in the environment caused an increase in the number of VRE cells, whereas the combination of vancomycin and dimer 5 caused a continuous decrease in CFUs of vancomycin-resistant cells with respect to total cells (Figure 7C). Therefore, these results confirm the ability of dimer 5 to protect VSE against the development of vancomycin resistance and also limit the growth of VRE cells. As conjugative transfer plays a role for the movement of the transposable element responsible for drug resistance,<sup>44</sup> we also checked the expression of the transposable element (Tn1546) in competitive inhibition study, as Tn1546 is a key player for drug resistance in VRE strains. We observed a >3-fold increase in Tn1546 expression in vancomycin-treated cells, whereas cells treated with the combination of vancomycin and dimer 5 showed the basal expression of the transposable element (Figure 7D). Therefore, these results demonstrate that bacteria gain vancomycin resistance due to the expression of the transposon element,

and dimer 5 inhibits the bacteria from developing this resistance.

The vanR/S two-component system regulates the expression of the vanHAX gene cluster responsible for vancomycin resistance (Figure 7E). Therefore, we investigated the impact of dimer 5 on vancomycin-mediated resistance. We quantified the change in expression of vanA operon genes, vanR, vanS, vanH, vanA, vanX, vanY, and vanZ, on the treatment of VRE with dimer 5 alone, vancomycin alone, and a combination of dimer 5 and vancomycin. Although vancomycin caused a multifold increase in expression of all the vanA operon genes (Figure 7F), the treatment of VRE with a combination of dimer 5 and vancomycin inhibited the vancomycin-mediated upregulation of vanA operon genes. Therefore, these results suggest that dimer 5 inhibits vancomycin-mediated activation of the two-component vanR/S system and does not allow the increase in expression of vanR, vanS, vanH, vanA, vanX, vanY, and vanZ genes. The reduced expression of these genes depicts the inhibition of conversion of D-Ala-D-Ala to D-Ala-D-Lac by affecting the vanR/S system. Therefore, the long-term



**Figure 8.** Scheme showing the mechanism of dimer 5 in inhibiting the emergence of vancomycin resistance.

treatment of VSE bacteria with the combination of dimer 5 and vancomycin does not allow the bacteria to develop drug resistance.

### 3. CONCLUSIONS

In summary, we present the synthesis of a class of cholic acid-derived dimeric amphiphiles where two cholic acid moieties were tethered through different alkylene chain spacers, and hydroxyl groups of cholic acid were conjugated to glycine or glycine-valine dipeptides. SAR studies showed that dimer 5 bearing glycine-valine dipeptides and a trimethylene spacer is the most effective against vancomycin-resistant bacteria. In-depth mechanistic studies revealed that dimer 5 can permeabilize the bacterial membranes and can generate ROS, causing a bactericidal effect. Apart from clearing the bacterial biofilms, topical application of dimer 5 did not cause any immunogenic reaction and cleared the VRE-mediated wound infections effectively. We further demonstrated that dimer 5 did not allow the bacteria to develop vancomycin resistance by inhibiting the expression of *vanA* operon-resistant genes and the Tn1546 transpose element (Figure 8). Therefore, the ability of the cholic acid-derived dimer amphiphiles to clear VRE infections and inhibit the emergence of vancomycin resistance makes them an ideal scaffold for future applications.

### 4. EXPERIMENTAL SECTION

**4.1. Materials.** Routine chromatography solvents, hydrochloric acid, sodium bicarbonate, and sodium chloride were purchased from Rankem, India, and used as such. Cholic acid, *n*-alkyl dibromides, CuBr, dicyclohexyl carbodiimide (DCC), anhydrous dichloromethane (DCM), anhydrous methanol, anhydrous *N,N'*-dimethyl formamide (DMF), and triethyl amine (TEA) were obtained from Sigma-Aldrich, USA. 1-Ethyl-3-(3-dimethylaminopropyl)carbodiimide hydrochloride (EDC-HCl), *N*-hydroxybenzotriazole (HOBt), *N,N*-diisopropylethyl amine (DIPEA), Boc-glycine, ninhydrin, and *n*-hexane were purchased from Spectrochem Pvt. Ltd., India. Propargyl amine was obtained from TCI Chemicals, and Boc-valine was purchased from Sisco Research Laboratories Pvt. Ltd., India. HPLC grade solvents were obtained and used as such from Thermo Fisher Scientific, USA. Tryptone soya (TS) broth (cat. #M011), brain heart infusion broth (BHIB) (cat. #M210), nutrient broth (cat. #M002), bacteriological agar powder (cat. #GRM026), and vancomycin hydrochloride (cat. #CMS21) were purchased from HiMedia, India. Trypsin–EDTA (cat. #S9418C), ciprofloxacin (cat. #17850), propidium iodide (cat. #P1470), 2',7'-dichlorofluorescein diacetate (cat. #35845), 3,3-diethylthiadicarbocyanine iodide (cat. #17375), and TRIzol reagent (cat. #T9424) were purchased from Sigma-Aldrich, USA. The LIVE/DEAD BacLight bacterial viability and counting kit was obtained from Invitrogen, USA. The cDNA Takara kit (cat. #6110A) and SYBR green (cat. #RR820A) were purchased from Takara Bio Inc., Japan. *S. aureus* (ATCC 25923), methicillin-resistant *S. aureus* (MRSA)

(ATCC 33591), vancomycin-susceptible *E. faecium* (VSE) (ATCC 19634), and vancomycin-resistant *E. faecium* (VRE) (ATCC 51559) were purchased from ATCC, USA.

**4.2. Methods.** Thin-layer chromatography was carried out on aluminum sheet coated with silica gel 60 GF 254 (Merck, Germany) and stained with ninhydrin and phosphomolybdic acid alcoholic solution. The  $^1\text{H}$  NMR spectra were recorded in  $\text{CDCl}_3$  and  $\text{DMSO}-d_6$  using a Bruker Avance 400 MHz spectrometer. Chemical shifts ( $\delta$ ) are reported in ppm using tetramethylsilane (TMS) as the internal standard. Mass spectrometry analysis was performed using a SCIEX TOF/TOF 5800 or MALDI-TOF. The purity of final molecules was checked with reverse phase HPLC using an Agilent HPLC system, equipped with a C18 column (S0005)  $4.6 \times 250$  mm ( $5 \mu\text{m}$ ) with UV detection at 230 nm. An HPLC isocratic 10 min run program was set using mixtures of solvent A (0.1% trifluoroacetic acid in water) and solvent B (0.1% trifluoroacetic acid in acetonitrile) at  $25^\circ\text{C}$  at a flow rate of 0.5 mL/min. The purity is >95%.

**4.3. Synthesis of Diazo Alkanes (Figure S1).** 1,3-Dibromopropane (100 mg, 1 equiv) was dissolved in dry DMF followed by addition of sodium azide (10 equiv). The reaction mixture was stirred at  $70^\circ\text{C}$  for 48 h. The formation of azide was monitored by TLC. The TLC plate was dipped into the solution of 20% triphenylphosphine in chloroform followed by ninhydrin staining. Formation of a purple color indicates the formation of azide. The reaction mixture was diluted with ethyl acetate and washed with brine ( $2 \times 200$  mL). The organic layer was dried over anhydrous sodium sulfate and concentrated by a rotary evaporator. Similarly, diazides of butane (10b), pentane (10c), and hexane (10d) were synthesized and characterized by  $^1\text{H}$  NMR.

**10a:** yield, 93%;  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ )  $\delta$ : 1.78–1.84 (m, 2H), 3.99 (t, 4H,  $J = 8$  Hz).

**10b:** yield, 95%;  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ )  $\delta$ : 1.52 (s, 4H), 3.41 (s, 4H).

**10c:** yield, 91%;  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ )  $\delta$ : 1.43–1.57 (m, 2H), 1.60–1.81 (m, 4H), 3.30–3.49 (m, 4H).

**10d:** yield, 92%;  $^1\text{H}$  NMR (400 MHz,  $\text{CDCl}_3$ )  $\delta$ : 1.32–1.39 (m, 4H), 1.57–1.75 (m, 4H), 3.23–3.44 (m, 4H).

**4.4. Synthesis of Compound 12.** Cholic acid (11) (2 g, 1 equiv), *N*-hydroxy benzotriazole (1.5 equiv), and EDC-HCl (1.5 equiv) were dissolved in 20 mL of anhydrous DMF, and DIPEA (3 equiv) was added to it. The reaction mixture was allowed to stir at room temperature for 15 min. Propargyl amine (3 equiv) was added drop-wise followed by stirring at room temperature for 24 h. The progress of reaction was monitored by TLC. The reaction mixture was diluted with DCM (250 mL) and washed with saturated sodium bicarbonate solution ( $3 \times 200$  mL) and brine ( $2 \times 200$  mL). The organic layer was dried over anhydrous sulfate and concentrated by a rotary evaporator. Compound 12 was purified by column chromatography on silica gel (230–400 mesh) using methanol/DCM (4.5% methanol in dichloromethane) as eluent to give a colorless solid. Yield, 92%;  $^1\text{H}$  NMR (400 MHz,  $\text{DMSO}-d_6$ )  $\delta$ : 0.68 (s, 3H), 0.83–2.22 (m, 32H), 3.07 (s, 1H), 3.08 (m, 1H), 3.60 (s, 1H), 3.77 (m, 1H), 3.80–3.82 (m, 2H), 4.02 (s, 1H), 4.03 (s, 1H), 4.11 (s, 1H), 4.33 (d, 1H,  $J = 4$  Hz), 8.22 (m, 1H).

**4.5. Synthesis of Compound 13.** Compound 12 (1.5 g, 1 equiv) was dissolved in 10 mL of anhydrous DCM. DMAP (6 equiv) and Boc-glycine (6 equiv) were added to the solution. The mixture was cooled to 0 °C, and a solution of DCC (6 equiv) in DCM (10 mL) was added dropwise. The reaction mixture was stirred at room temperature for 4 h. After completion of the reaction, it was cooled to 0 °C and filtered to remove DCU. The filtrate was diluted with DCM (250 mL) and washed with saturated sodium bicarbonate solution (2 × 200 mL) and brine (2 × 200 mL). The organic phase was dried over anhydrous sodium sulfate, and the solvent was removed under vacuum. The crude mixture was purified by column chromatography on silica gel (230–400 mesh) using ethyl acetate/petroleum ether (35% ethyl acetate in petroleum ether) as eluent to give a white solid. Yield, 83%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.75 (s, 3H), 0.83 (d, 3H), 0.85 (s, 3H) 0.90–2.26 (m, steroid), 3.90–3.96 (m, 8H), 4.66 (m, 1H), 5.02–5.22 (m, 4H), 6.33 (s, 1H).

**4.6. Synthesis of Compounds 14–17.** Compound 13 (500 mg, 2.2 equiv) was dissolved in anhydrous methanol, and triethylamine (2.2 equiv) was added to it, followed by CuBr (0.1 equiv) addition in the dark. Different diazidoalkanes (1 equiv) were added to the reaction mixture to yield corresponding molecules. The reaction mixture was allowed to stir at room temperature under dark conditions for 72 h. After completion of the reaction, the solvent was evaporated using a rotary evaporator. The crude mixture was diluted with ethyl acetate and washed with saturated ammonium acetate (3 × 200 mL) to remove copper traces. The organic layer was washed with brine (2 × 200 mL) and dried over sodium sulfate, and the solvent was removed by a rotary evaporator. The crude mixture was purified by CombiFlash chromatography using methanol/DCM (4–5% methanol in DCM) as the mobile phase.

Compound 14: yield, 71%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.74 (s, 6H), 0.79–2.26 (m, 124H), 3.86–4.15 (m, 12H), 4.35 (bs, 3H), 4.49 (bs, 3H), 4.62 (s, 2H), 5.02–5.09 (m, 2H), 5.19 (s, 3H), 5.32 (s, 1H), 6.71 (s, 1H), 7.51 (s, 2H).

Compound 15: yield, 74%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.74 (s, 6H), 0.80–2.28 (m, 128H), 3.91–4.02 (m, 11H), 4.32 (t, 3H, *J* = 8 Hz), 4.50 (bs, 3H), 4.63 (s, 2H), 5.1 (s, 2H), 5.09 (bs, 1H), 5.29 (bs, 1H), 6.85 (s, 1H), 7.50 (s, 2H).

Compound 16: yield, 78%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.79 (s, 6H), 0.80–2.26 (m, 134H), 3.86–4.15 (m, 13H), 4.35 (s, 3H), 4.50 (s, 3H), 4.62 (s, 2H), 5.02 (s, 2H), 5.19 (s, 3H), 5.32 (s, 1H), 6.71 (s, 1H), 7.51 (s, 2H).

Compound 17: yield, 77%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.73 (s, 6H), 0.80–2.26 (m, 148H), 3.89–4.22 (m, 11H), 4.29–4.32 (m, 3H), 4.47–4.49 (d, 3H, *J* = 8 Hz), 4.6–4.7 (bs, 2H), 5.0 (s, 1H), 5.17–5.22 (m, 4H), 5.30 (s, 2H), 6.90 (s, 1H), 7.50 (s, 2H).

**4.7. Synthesis of Dimers 1–4.** The Boc-protected dimer molecules (14–17) (300 mg) were dissolved in anhydrous DCM (10 mL). The mixture was cooled to 0 °C, and 4 M HCl in dioxane (2 mL) was added to it dropwise. After 30 min, the reaction mixture was shifted to room temperature and stirred for 1 h. After completion of the reaction, the solvent was evaporated by a rotary evaporator, and the compound was precipitated using *n*-hexane. Repeated washes of *n*-hexane were given to obtain pure molecules.

Dimer 1: yield, 72%; <sup>1</sup>H NMR (400 MHz, DMSO-*d*<sub>6</sub>) δ: 0.73 (s, 6H), 0.75–2.10 (m, 77H), 3.67–3.80 (m, 6H), 3.87–4.01 (m, 6H), 4.22 (bs, 4H), 4.33 (bs, 4H), 4.58 (bs, 2H), 4.91 (s, 2H), 5.12 (s, 2H), 7.89 (s, 2H), 8.30 (bs, 2H), 8.52 (bs, 18H). HRMS (MALDI/TOF) calculated for (C<sub>69</sub>H<sub>111</sub>N<sub>14</sub>O<sub>14</sub>)<sup>+</sup> [*M* + *H*]<sup>+</sup>: 1359.8398; observed: 1359.8396; HPLC: 1.792 min (>99%).

Dimer 2: yield, 75%; <sup>1</sup>H NMR (400 MHz, DMSO-*d*<sub>6</sub>) δ: 0.66 (s, 6H), 0.73–2.10 (m, 82H), 3.67–3.80 (m, 6H), 3.87–4.01 (m, 6H), 4.21 (s, 5H), 4.55 (bs, 2H), 4.88 (s, 2H), 5.10 (s, 2H), 7.89 (s, 2H), 8.36 (s, 2H), 8.60 (s, 18H). HRMS (MALDI/TOF) calculated for (C<sub>70</sub>H<sub>113</sub>N<sub>14</sub>O<sub>14</sub>)<sup>+</sup> [*M* + *H*]<sup>+</sup>: 1373.8555; observed: 1373.853; HPLC: 1.789 min (>99%).

Dimer 3: yield, 73%; <sup>1</sup>H NMR (400 MHz, DMSO-*d*<sub>6</sub>) δ: 0.67 (s, 6H), 0.74–2.11 (m, 81H), 3.86–4.05 (m, 8H), 4.22 (s, 4H), 4.50 (bs, 2H), 4.90 (s, 2H), 5.11 (s, 2H), 7.89 (s, 2H), 8.34 (s, 2H), 8.56 (s,

18H). HRMS (MALDI/TOF) calculated for (C<sub>71</sub>H<sub>115</sub>N<sub>14</sub>O<sub>14</sub>)<sup>+</sup> [*M* + *H*]<sup>+</sup>: 1387.8711; observed: 1387.8716; HPLC: 1.786 min (>99%).

Dimer 4: yield, 58%; <sup>1</sup>H NMR (400 MHz, DMSO-*d*<sub>6</sub>) δ: 0.70 (s, 6H), 0.76–2.13 (m, 72H), 3–3.74 (m, 6H), 3.89–4.22 (m, 6H), 4.24 (d, 4H, *J* = 8 Hz), 4.35 (s, 4H), 4.94 (s, 2H), 5.14 (s, 2H), 7.90 (s, 2H), 8.53 (s, 18H). HRMS (MALDI/TOF) calculated for (C<sub>72</sub>H<sub>117</sub>N<sub>14</sub>O<sub>14</sub>)<sup>+</sup> [*M* + *H*]<sup>+</sup>: 1401.8868; observed: 1401.8852; HPLC: 1.767 min (>99%).

**4.8. Synthesis of Compound 18.** Compound 13 (1.5 g) was dissolved in anhydrous DCM (10 mL). The mixture was cooled to 0 °C, and 4 M HCl in dioxane (2 mL) was added to it dropwise. The reaction mixture was shifted to room temperature after 30 min and stirred for 1 h. After completion of deprotection, the excess of solvent was evaporated by a rotary evaporator, and the compound was precipitated with *n*-hexane. Yield, 89%; <sup>1</sup>H NMR (400 MHz, DMSO-*d*<sub>6</sub>) δ: 0.71 (s, 3H), 0.80–2.33 (m, 43H), 3.10–3.11 (m, 2H), 3.75–3.81 (m, 6H), 3.82–3.95 (m, 4H), 4.62 (bs, 1H), 4.94 (s, 1H), 5.14 (s, 1H), 8.29 (s, 1H), 8.48 (s, 9H).

**4.9. Synthesis of Compound 19.** Compound 18 (800 mg) was dissolved in anhydrous DCM (10 mL), and DIPEA (6 equiv) was added. To this mixture, EDC-HCl (6 equiv) and HOBt (6 equiv) were added, and the reaction mixture was stirred at room temperature for 15 min. Boc-valine (6 equiv) was added to the reaction mixture and allowed to stir overnight. The progress of reaction was monitored with TLC. The mixture was diluted with DCM (200 mL) and washed with saturated NaHCO<sub>3</sub> solution (2 × 200 mL) and brine (2 × 200 mL). The organic phase was dried over anhydrous sodium sulfate, and the solvent was reduced under vacuum. The crude mixture was purified by column chromatography on silica gel (230–400 mesh) using methanol/DCM (3% methanol in DCM) as eluent to give a white solid. Yield, 87%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.74 (s, 3H), 0.82–2.22 (m, 84H), 3.99–4.11 (m, 10H), 5.19 (bs, 1H), 5.19–5.50 (m, 5H), 6.75 (bs, 1H), 7.02 (bs, 1H), 7.22 (bs, 1H).

**4.10. Synthesis of Compounds 20–23.** Compound 19 (500 mg, 2.2 equiv) was dissolved in anhydrous methanol, and triethylamine (2 equiv) was added to it, followed by CuBr (0.1 equiv) in dark conditions. Different diazidoalkanes (1 equiv) were added to the mixture to yield corresponding molecules. The reaction mixture was allowed to stir at room temperature under dark conditions for 72 h. After completion of the reaction, the solvent was evaporated using a rotary evaporator. The crude mixture was diluted with ethyl acetate and washed with saturated ammonium acetate (3 × 200 mL) to remove copper traces and then with brine (2 × 200 mL). The organic layer was dried over sodium sulfate, and the solvent was reduced by a rotary evaporator. The crude mixture was purified by CombiFlash chromatography using methanol/DCM (5–7% methanol in DCM) as the mobile phase.

Compound 20: yield, 76%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.73 (s, 6H), 0.80–2.11 (m, 164H), 3.96–4.28 (m, 15H), 4.35–4.38 (m, 3H), 4.50–4.65 (m, 5H), 5.03 (s, 2H), 5.16 (s, 2H), 5.32 (s, 3H), 7.01 (s, 2H), 7.60 (s, 2H).

Compound 21: yield, 72%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.73 (s, 6H), 0.78–2.20 (m, 166H), 3.90–4.17 (m, 16H), 4.35 (s, 4H), 4.49–4.62 (m, 5H), 5.03 (s, 2H), 5.16 (s, 2H), 5.23–5.50 (m, 3H), 5.63 (bs, 1H), 7.02 (bs, 2H), 7.54 (s, 2H).

Compound 22: yield, 78%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.73 (s, 6H), 0.78–2.17 (m, 148H), 4.09–4.15 (m, 14H), 4.32 (t, 3H, *J* = 8 Hz), 4.49 (s, 3H), 4.57 (s, 2H), 5.01 (s, 2H), 5.16 (s, 2H), 5.35 (s, 2H), 7.02 (bs, 2H), 7.53 (s, 2H).

Compound 23: yield, 73%; <sup>1</sup>H NMR (400 MHz, CDCl<sub>3</sub>) δ: 0.73 (s, 6H), 0.78–2.17 (m, 152H), 4.09–4.15 (m, 14H), 4.32 (t, 3H, *J* = 8 Hz), 4.49 (s, 3H), 4.57 (s, 2H), 5.01 (s, 2H), 5.16 (s, 2H), 5.35 (s, 2H), 7.02 (bs, 2H), 7.53 (s, 2H).

**4.11. Synthesis of Dimers 5–8.** The Boc-protected dimer molecules (20–23) (300 mg) were dissolved in anhydrous DCM (10 mL) in different round-bottom flasks. The mixtures were cooled to 0 °C, and 4 M HCl in dioxane (2 mL) was added to it dropwise. After 30 min, the reaction mixtures were shifted to room temperature and stirred for 1 h. After completion of deprotection, the solvent was evaporated by a rotary evaporator, and compounds were precipitated



using *n*-hexane. Repeated washes of *n*-hexane were given to obtain pure molecules.

Dimer 5: yield, 78%;  $^1\text{H}$  NMR (400 MHz,  $\text{DMSO}-d_6$ )  $\delta$ : 0.68 (s, 6H), 0.74–2.15 (m, 100H), 2.31–2.40 (m, 2H), 3.70 (s, 7H), 3.73–3.79 (m, 4H), 3.82–4.07 (m, 7H), 4.25 (s, 3H), 4.34–4.37 (m, 3H), 4.50 (bs, 2H), 4.84 (s, 2H), 5.06 (s, 2H), 7.97 (s, 2H), 8.33–8.40 (m, 18H), 8.06–8.90 (m, 1H), 9.11–9.30 (m, 5H). HRMS (MALDI/TOF) calculated for  $(\text{C}_{99}\text{H}_{166}\text{N}_{20}\text{O}_{20})^+ [\text{M} + \text{H}]^+$ : 1995.2576; observed: 1995.2578; HPLC: 1.784 min (>99%).

Dimer 6: yield, 74%;  $^1\text{H}$  NMR (400 MHz,  $\text{DMSO}-d_6$ )  $\delta$ : 0.67 (s, 6H), 0.72–2.15 (m, 105H), 3.91–4.05 (m, 11H), 4.22 (s, 3H), 4.34 (s, 3H), 4.55 (bs, 1H), 4.82 (s, 2H), 5.04 (s, 2H), 7.90 (s, 2H), 8.25–8.33 (m, 18H), 8.85–8.90 (m, 1H), 9.11–9.30 (m, 5H). HRMS (MALDI/TOF) calculated for  $(\text{C}_{100}\text{H}_{168}\text{N}_{20}\text{O}_{20})^+ [\text{M} + \text{H}]^+$ : 1969.2732; observed: 1969.2735; HPLC: 1.772 min (>99%).

Dimer 7: yield, 76%;  $^1\text{H}$  NMR (400 MHz,  $\text{DMSO}-d_6$ )  $\delta$ : 0.67 (s, 6H), 0.73–2.15 (m, 105H), 3.69–3.80 (m, 11H), 3.98–4.10 (m, 8H), 4.23 (s, 3H), 4.24–4.31 (m, 3H), 4.50 (s, 1H), 4.83 (s, 2H), 5.05 (s, 2H), 7.84 (s, 2H), 8.23–8.31 (m, 18H), 8.80 (s, 1H), 8.82–9.30 (m, 5H). HRMS (MALDI/TOF) calculated for  $(\text{C}_{101}\text{H}_{169}\text{N}_{20}\text{O}_{20})^+ [\text{M} + \text{H}]^+$ : 1982.2816; observed: 1982.2818; HPLC: 1.747 min (>99%).

Dimer 8: yield, 72%;  $^1\text{H}$  NMR (400 MHz,  $\text{DMSO}-d_6$ )  $\delta$ : 0.67 (s, 6H), 0.72–2.14 (m, 109H), 3.83–4.05 (m, 10H), 4.22–4.29 (m, 6H), 4.50 (s, 2H), 4.82 (s, 2H), 5.04 (s, 2H), 7.89 (s, 2H), 8.20–8.40 (m, 18H), 8.80–8.90 (m, 1H), 9.01–9.30 (m, 5H). HRMS (MALDI/TOF) calculated for  $(\text{C}_{102}\text{H}_{171}\text{N}_{20}\text{O}_{20})^+ [\text{M} + \text{H}]^+$ : 1996.2973; observed: 1996.2975; HPLC: 1.776 min (>99%).

**4.12. Antibacterial Activity.** Minimum inhibitory concentrations ( $\text{MIC}_{99}$ ) against different strains were determined according to the guidelines of the Clinical Laboratory Standards Institute (CLSI).<sup>45</sup> *S. aureus*, methicillin-resistant *S. aureus* (MRSA), vancomycin-susceptible *E. faecium* (VSE), and vancomycin-resistant *E. faecium* (VRE) cells were cultured overnight in their respective media, tryptone soya broth (TSB), nutrient broth (NB), tryptone soya broth (TSB), and brain heart infusion broth (BHIB) media, respectively. The log phase-cultured bacterial cells were subjected to centrifugation (5000 rpm for 10 min) and resuspended in fresh media to get  $10^6$  cells/mL. All amphiphiles were serially diluted from 128 to 0.25  $\mu\text{g}/\text{mL}$  (100  $\mu\text{L}$ ) in 96-well plates in the respected media. One hundred microliters of bacterial suspension was added in each well of a 96-well plate. Finally, plates were incubated at 37 °C for 14–16 h, and OD was observed at 600 nm using a Spectramax M5, a multimode microplate reader (Molecular Devices, Sunnyvale, CA, USA). Ciprofloxacin and vancomycin were used as a positive control, and we represented antibacterial activity as  $\text{MIC}_{99}$ .

**4.13. Bactericidal Time-Kill Kinetic Study.** VRE cells were grown up to the log phase in BHIB media and centrifuged at 5000 rpm for 10 min. Cells were then resuspended in fresh media to  $10^6$  cells/mL and treated with dimer 5 at 4, 8, and 16  $\mu\text{g}/\text{mL}$  (1 $\times$ , 2 $\times$ , and 4 $\times$   $\text{MIC}_{99}$ ) for different time points (0, 1, 3, 6, 12, and 24 h) at 37 °C. After each time point, untreated and treated suspension cells were serially diluted and plated on BHIB agar plates. Plates were incubated at 37 °C for 16–18 h. Colony-forming units were counted, and bacterial burden was expressed as  $\log_{10}$  CFU/mL as a function of time.

**4.14. Membrane Potential Fluorescence Assay.** VRE cells were grown until the log phase, and cells were harvested by centrifugation at 5000 rpm for 10 min. The harvested cells were resuspended in PBS (PBS with 1% glucose) to  $\sim 10^7$  cells/mL. These cells were stained with  $\text{DiSC}_2(\text{S})$  (100  $\mu\text{M}$ ) dye and incubated at 37 °C. After every 10–15 min, fluorescence at 670 nm with an excitation at 637 nm was measured until it got saturated. Next, KCl (100 mM) was added to stabilize the fluorescence of suspended cells. We treated saturated cells with different concentrations of dimer 5. After treatment, we incubated samples at 37 °C for 15 min, transferred the suspension into the 96-well plate, and set up the kinetics in a Spectramax M5 multimode microplate reader (Molecular Devices, Sunnyvale, CA, USA) for 30 min.

**4.15. Propidium Iodide Uptake Assay.** VRE cells were harvested at the log phase after centrifugation at 5000 rpm for 10 min. Cells were diluted to  $10^6$  cells/mL in PBS and treated with different concentrations of dimer 5 (4, 8, and 16  $\mu\text{g}/\text{mL}$ ) for 1 h. Further, cells were stained with PI (10  $\mu\text{g}/\text{mL}$ ) and again incubated for 10 min. Incubated cells were washed twice with PBS and finally resuspended in 200  $\mu\text{L}$  of PBS. Resuspended samples were analyzed by flow cytometry using a BD FACS Verse flow cytometer (BD Biosciences, San Jose, CA) under a propidium iodide channel.

**4.16. ROS Generation Assay.** VRE cells were cultured in BHIB media at 37 °C for 14–16 h. The culture was diluted in a 1:100 ratio and grown up to an OD of 0.5. These cells were harvested at 5000 rpm for 10 min, washed with PBS (pH 7.4), and treated with DCFH-DA (100  $\mu\text{M}$ ) for 1 h at 37 °C. DCFH-DA-stained cells were treated with an increasing concentration of dimer 5 for 15 min and washed twice with PBS at 5000 rpm for 5 min. The supernatant was added to a 96-well plate, and fluorescence was measured at 535 nm using excitation at 485 nm in a Spectramax M5 multimode microplate reader (Molecular Devices, Sunnyvale, CA, USA).

**4.17. Planktonic Cell Live/Dead Staining.** VRE cells were grown up to the log phase and harvested at 5000 rpm for 10 min. These cells were further diluted in PBS to  $10^6$  cells/mL and treated with dimer 5 at its  $\text{MIC}_{99}$  for 1 h. Both untreated and treated cells were stained with SYTO9 and PI of the LIVE/DEAD BacLight bacterial viability and counting kit. SYTO9 and PI dyes were mixed into a 1:1 ratio, and 1  $\mu\text{L}$  of this mixture was used to stain the cells for 15 min. After incubation, VRE cells were washed with PBS and finally resuspended into 50  $\mu\text{L}$  of PBS. The suspension (10  $\mu\text{L}$ ) was put on the glass slide and covered with a cover glass. After sealing the dried cover glass with transparent nail enamel, these slides were observed under an inverted confocal microscope (Leica TCS SP8, Leica Germany) under a 63 $\times$  oil objective in fluorescein isothiocyanate (FITC for SYTO9, green) and tetramethylrhodamine isocyanate (TRITC for PI, red) channels.

**4.18. Transmission Electron Microscopy.** Log phase-cultured VRE cells were harvested and diluted in PBS to get  $10^6$  cells/mL. These cells were treated with dimer 5 for 1, 2, and 6 h. Untreated and treated cells were then fixed using a mixture of 4% PFA (paraformaldehyde) and 2.5% glutaraldehyde for 4 h at room temperature. These fixed cells were then dehydrated using an ethanol gradient. Dehydrated cells were embedded in Epon 812 resin, and ultrathin sections were stained using a mixture of lead citrate and uranyl acetate. Imaging of sections was done on a JEM-1400 Flash Electron Microscope, JEOL Ltd., Japan. Microscopic image analysis was done using different fields of view manually.

**4.19. Biofilm Degradation Assay.** VRE and *S. aureus* cells were inoculated and cultured in TSB media at 37 °C for 14 h. The bacterial suspension was diluted to  $2 \times 10^6$  CFU/mL in TSB with 1.5% glucose and added into a sterile six-well plate with heat sterile coverslips (18  $\times$  18 mm). These plates were incubated at 37 °C for 7–8 days for VRE biofilms and 4 days for *S. aureus* biofilms. After every 48 h, media were replenished with fresh TSB media. We treated biofilms at different concentrations of dimer 5 for different time points (0, 3, 6, and 12 h). Untreated and treated biofilms were washed with PBS and treated with trypsin–EDTA (100  $\mu\text{L}$ ) for 10 min. Finally, the resuspension was serially diluted and plated on agar plates, and plates were incubated for 18–24 h at 37 °C. The viability of VRE and *S. aureus* bacterial cells within the biofilms was accessed by the colony-forming units (CFUs) at different time points. A set of biofilms after 12 h treatment with dimer 5 were stained with the LIVE/DEAD BacLight bacterial viability and counting kit and imaged using a confocal microscope (Leica TCS SP8; Leica, Germany).

**4.20. Flow Cytometry.** BALB/c male mice (average weight,  $\sim 20$  g; age,  $\sim 6$  weeks) were obtained from the institute's small animal facility. The dorsal fur of BALB/c male mice (average weight,  $\sim 20$  g; age,  $\sim 6$  weeks) was shaved, and the selected area was cleaned with Veet cream. We left the naked area open for 2 days to avoid any inflammatory reaction due to Veet cream. The mice were divided into three groups ( $n = 6/\text{group}$ ) where we applied saline once a day to group 1 mice, imiquimod cream (3.125 mg/mice) once a day to



group 2 mice, and dimer 5 at 40 mg/kg thrice a day to group 3 mice at the dorsal open surface of mice for 6 days. On day 7, mice were euthanized, and skin was isolated in a tube containing cold 1× PBS. Infiltration of monocytes, macrophages, and neutrophils was quantified by flow cytometry using a previously published protocol.<sup>46</sup>

**4.21. Wound Infection Model.** BALB/c male mice (average weight, ~20 g; age, ~6 weeks) were used from the institute's small animal facility. BALB/c male mice were anesthetized by peritoneal injection of ketamine and xylazine (100  $\mu$ L/mice). The dorsal fur of mice was shaved and cleaned with Veet cream, and the naked skin area was made sterile with povidone iodine solution.<sup>47</sup> An open wound of ~1 cm<sup>2</sup> in diameter was created with a scissor, and 20  $\mu$ L of VSE suspension (~10<sup>6</sup> cells) was applied on the open wound. After 18 h of infection, mice were divided into three groups ( $n$  = 6/group). First group mice ( $n$  = 6) were treated with saline (thrice a day) topically, whereas second group ( $n$  = 6) mice were treated with vancomycin (40 mg/kg, thrice a day) topically, third group mice ( $n$  = 6) were treated with dimer 5 (40 mg/kg, thrice a day) topically, and treatment was continued for 3 days. We sacrificed all the mice and quantified bacterial burden by CFU analysis. In the second experiment, we used VRE cells for infection, and infected mice were divided into four groups ( $n$  = 6/group) with group 1 of untreated mice, group 2 of vancomycin-treated mice, group 3 of dimer 5-treated mice and group 4 of ciprofloxacin-treated mice. In the third experiment, we used *S. aureus* cells for the infection, and infected mice were divided into three groups ( $n$  = 6/group) with group 1 of untreated mice, group 2 of neomycin-treated mice, and group 3 of dimer 5-treated mice.

**4.22. Drug Resistance Studies.** VRE cells were cultured in BHIB media at 37 °C for 12–14 h and diluted to 2 × 10<sup>6</sup> CFU/mL as a working suspension. Cells were treated with dimer 5 and ciprofloxacin in a 96-well plate from 128 to 0.25  $\mu$ g/mL at 37 °C for 24 h to determine the MIC<sub>99</sub>. The next day, the bacterial suspension from the sub-MIC<sub>99</sub> well was diluted 100 times and added into a new plate for another cycle of bacteria–drug incubation. The cells were continuously passaged until 56 generations, and the MIC<sub>99</sub> of dimer 5 and ciprofloxacin was identified. In the next set of experiments, we used VSE cells and cultured them in the presence of vancomycin, dimer 5, and a combination of vancomycin with 2 and 8  $\mu$ g/mL dimer 5. The VSE cells were continuously passaged until 110 generations, and the MIC<sub>99</sub> of each group was measured.

**4.23. Competitive Inhibition Study.** We inoculated and cultured VRE and VSE cells in TSB media at 37 °C for 14–16 h. Cultured cells were diluted to 1:100, further grown at 37 °C for 9–10 h, and then mixed in a 1:1 ratio (VRE:VSE).<sup>20</sup> The mixture (VRE and VSE) of cells was diluted (1:8) with media and grown in three sets, untreated control, vancomycin treatment of 2  $\mu$ g/mL, and combination treatment of vancomycin (2  $\mu$ g/mL) and dimer 5 (2  $\mu$ g/mL) at sub-inhibitory concentration. The untreated and treated cells were incubated for 24 h at 37 °C, and around ~10<sup>7</sup> cells were serially diluted and plated on vancomycin (6  $\mu$ g/mL) plates that will allow only vancomycin-resistant cells to grow on the plate. The cells were continuously cultured in all three sets until seven passages.

**4.24. Gene Expression Studies.** VRE cells were grown until the log phase in BHIB at the sub-inhibitory concentration of vancomycin (2  $\mu$ g/mL), dimer 5 (2  $\mu$ g/mL), and the combination of vancomycin and dimer 5 (2  $\mu$ g/mL each) at 37 °C. The total RNA was extracted using the TRIzol reagent and quantified at the absorbance of 260/280 nm ratio using a NanoDrop spectrophotometer (Thermo Fisher Scientific, USA). In addition, for evaluating the expression of the Tn1546 transposable element in competitive inhibition study, we extracted the total RNA from untreated, vancomycin-treated, and combination-treated cells at the end of 7th day. Complementary DNA (cDNA) was prepared using 2  $\mu$ g of RNA from the cDNA reverse transcriptase kit (Takara Bio, CA, USA). The mRNA levels of *vanR*, *vanS*, *vanH*, *vanA*, *vanX*, *vanY*, *vanZ*, and Tn1546 were quantified relative to the control genes (Gyrase B) in a qRT-PCR machine (ABI-Quant studio 6 Real Time PCR system, Thermo Fisher Scientific, Singapore). Thermal cycling was performed by a two-step PCR amplification standard procedure with 95 °C for 15 s, then 40 cycles

of 95 °C for 5 s, and 60 °C for 20 s. The expression data were collected and further processed to calculate the 2<sup>−ΔΔCT</sup>.<sup>48</sup> Primers used are mentioned in Table S1.

**4.25. Statistical Analysis.** Statistical analysis was performed using two-tailed unpaired Student's *t*-test to compare differences between two different groups. One-way or two-way ANOVA was performed to compare multiple groups. Statistical analysis was performed using GraphPad Software (Prism 6). \**p* < 0.05, \*\**p* < 0.01, \*\*\**p* < 0.001, and \*\*\*\**p* < 0.0001.

**4.26. Ethical Statement.** All animal experiments were performed after the due approval of the Institutional Animal Ethical Committee of RCB (RCB/IAEC/2016/053) as per the guidelines of Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA), India.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.2c01293>.

Primer sequences used for the study, schematic showing the synthesis of diazidoalkanes, fluorescence micrographs of VRE bacteria after staining with SYTO/PI, transmission electron micrographs, SYTO/PI-stained fluorescence micrographs of *S. aureus* biofilms, time-dependent change in CFUs on treatment of VRE biofilms, gating strategy for immune cells, and characterization of amphiphiles (PDF)

Molecular formula strings (CSV)

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## Notes

The authors declare no competing financial interest.

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## ABBREVIATIONS

AMPs, antimicrobial peptides; BHIB, brain heart infusion broth; CAP, cholic acid–peptide conjugate; CFUs, colony-forming units; DCFH-DA, 2',7'-dichlorofluorescein diacetate; DCU, dicyclohexylurea; DIPEA, *N,N*-diisopropylethylamine; DiSC<sub>2</sub>(S), 3',3'-diethylthiadicarbocyanine; EDC.HCl, *N*-(3-dimethylaminopropyl)-*N'*-ethylcarbodiimide hydrochloride; FITC, fluorescein isothiocyanate; HOBt, hydroxybenzotriazole; IMQ, imiquimod; LLC, Lewis lung carcinoma; NB, nutrient broth; PI, propidium iodide; SLAY, surface-localized antimicrobial display; TEA, triethylamine; TEM, transmission electron microscopy; TRITC, tetramethylrhodamine isothiocyanate; TSB, tryptone soya broth; VISA, vancomycin intermediate-sensitive *Enterococci*; VRSA, vancomycin-resistant *S. aureus*; VSE, vancomycin-sensitive *Enterococci*

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